Feature Engineering Report

1. Introduction

The Economic Intelligence Framework developed in this study integrates heterogeneous datasets from multiple international and domestic sources. These datasets differ in reporting frequency, temporal coverage, and economic significance. Consequently, feature engineering extends beyond creating new variables and includes data harmonisation, temporal alignment, cross-dataset integration, lag construction, and transformation of variables into a common monthly forecasting dataset.

The final forecasting dataset consists of **276 monthly observations**, four predictor variables, and one target variable. These engineered features form the common input for both the SARIMAX statistical model and the LSTM deep learning model, although each model represents temporal dependencies differently.

2. Cross-Dataset Integration Strategy

The forecasting framework integrates five independent economic datasets into a unified monthly analytical dataset representing the following variables.

Each dataset underwent independent cleaning before integration. Date formats were standardised into a common **Year–Month (YYYY-MM)** representation to ensure temporal consistency across all sources. Standardising the temporal key enabled reliable merging, preserved chronological ordering, and ensured that every observation represented the same economic period.

After harmonisation, the datasets were merged using the monthly timestamp as the primary key. Monthly datasets were merged directly, while datasets reported at other frequencies underwent additional preprocessing before integration.

The Human Capital Project dataset was intentionally excluded from the forecasting dataset because its annual reporting frequency is incompatible with monthly forecasting. Instead, it forms a complementary socioeconomic component within the broader Economic Intelligence Framework.

3. Daily-to-Monthly Aggregation of the Baltic Dry Index

Because the forecasting framework operates at monthly resolution, the BDI required temporal aggregation before integration. Rather than averaging daily values directly, engineered monthly indicators were created to preserve economically meaningful information regarding shipping market behaviour. The following features were derived: monthly standard deviation of daily BDI values (BDI_std) and monthly BDI returns (monthly_return_bdi).

Using engineered monthly indicators rather than raw daily observations offers several advantages: eliminates temporal misalignment; reduces high-frequency noise; and preserves information about shipping market dynamics;

These variables provide complementary representations of international logistics markets, with **BDI_std** capturing market uncertainty and **monthly_return_bdi** representing directional changes in shipping costs.

The monthly standard deviation (**BDI_std**) was selected because it quantifies shipping market volatility throughout each month, allowing the forecasting models to capture periods of heightened uncertainty within international logistics markets. Similarly, **monthly_return_bdi** measures the relative month-to-month change in shipping costs, providing information regarding shipping market momentum rather than absolute freight prices. This aggregation strategy reduces high-frequency noise while preserving the underlying economic signals most relevant to medium-term food inflation forecasting.

4. Justification of Predictor Variables

Each predictor variable was selected based on its theoretical relationship with food inflation and its ability to represent a distinct dimension of Botswana's macroeconomic environment.

Brent_USD_per_barrel

Brent crude oil prices represent global energy market conditions.

Energy costs influence nearly every stage of the food supply chain, including agricultural production, fertiliser production, transportation and distribution. Changes in oil prices therefore transmit through the economy and ultimately influence domestic food prices.

policy_rate

The Botswana policy rate represents the principal domestic monetary policy instrument. Changes in the policy rate affect borrowing costs, household expenditure and inflation expectations. Including the policy rate enables the forecasting models to capture domestic macroeconomic conditions alongside international market influences. From exploratory analysis, the policy rate is constant for long periods of time. However, we expect it to have delayed effects, so we examine it as a lagged feature in the classical model.

BDI_std

BDI_std measures the monthly volatility of international shipping markets. Periods of elevated shipping volatility often correspond to supply-chain disruptions, increased transportation uncertainty, and changing freight demand. These disruptions may subsequently influence import costs and domestic food prices.

monthly_return_bdi

Monthly_return_bdi measures the month-to-month percentage change in shipping costs.

Whereas BDI_std represents market stability, monthly_return_bdi captures shipping market momentum and directional change. Including both variables enables the forecasting framework to distinguish between persistent market instability and changing transportation costs.

Target Variable (FAO_23014)

The forecasting target is **FAO_23014**, representing the monthly Food Price Index that the forecasting models seek to predict.

5. Merge Strategy

Following preprocessing and harmonisation, all datasets were integrated using a structured merge strategy, which involved aligning on a common monthly timeline and merging using the year_month timestamp.

The integration workflow follows a deterministic merge strategy in which the FAO monthly dataset serves as the reference timeline. This approach guarantees that each observation within the forecasting dataset represents a complete macroeconomic snapshot for a single calendar month.

6. Lag Structures

Economic variables rarely affect food inflation immediately. Instead, their influence often appears after a measurable delay. Consequently, lag structures were incorporated into the forecasting framework.

SARIMAX Model

The SARIMAX model employs **explicitly engineered lag features** selected through statistical diagnostics. Cross-Correlation Function (CCF) analysis was first used to identify the temporal delay at which each predictor exhibited the strongest relationship with food inflation. These lag structures were subsequently validated using Granger causality testing. After these tests, we included the following as exogenous variables: BDI_std_lag7, policy_rate_lag12 and Brent_lag16. The original policy_rate was excluded from the forecast due to high multicollinearity with policy_rate_lag12. The original Brent_USD_per_barrel was also excluded because it did not show any significant relationship at lag 0.

Cross-Correlation Function analysis did not identify a statistically significant delayed relationship between monthly BDI returns and food inflation (even at lag 0), and subsequent Granger causality testing likewise failed to demonstrate meaningful predictive power at the examined lag intervals. Consequently, monthly_return_bdi was dropped from the analysis.

In addition to lags, the team also accounted for seasonality and structural breaks. A seasonal decomposition analysis was conducted in addition to a Bai-Perron test for structural breaks. Evidence of seasonality was found in both the dependent and independent variables. Structural breaks were also identified that coincided with historical events such as the 2008 recession, COVID-19 pandemic and 2022 Ukraine War. To handle seasonality and structural breaks, dummy variables were created: month_2 to month_12 for seasonality and break_2007, break_2009 and break_2022 for the structural breaks. Month_1 was excluded to avoid multicollinearity. Break_2007 was also dropped due to high multicollinearity with break_2009. For simplicity, dummy variables were only created for the dependent variable.

These lagged predictors and seasonality and structural break dummy variables were incorporated as exogenous regressors within the final ARIMA(0,1,2) model.

LSTM Model

Unlike ARIMAX, the LSTM does not require manually engineered lag variables. Instead, each prediction is generated using the previous **twelve consecutive months** of observations.

Each training sample therefore consists of: twelve months of historical observations; four predictor variables; and one prediction for the following month.

The recurrent architecture automatically learns temporal relationships from these sequences through its internal memory mechanism.

7. Feature Transformations

Several transformations were performed before model training.

Chronological Ordering

All observations were sorted chronologically before preprocessing to preserve temporal dependencies.

Temporal Harmonisation

Variables originating from different reporting frequencies were transformed into monthly observations.

Sequence Transformation

For the LSTM model, overlapping twelve-month sliding windows were generated from the monthly dataset.

Each sequence has dimensions: **(12 × 6)** representing twelve monthly observations across six predictor variables.

8. Feature Scaling

Because the predictor variables possess different numerical ranges, feature scaling was performed before training the LSTM model.

Separate scaling procedures were applied to: predictor variables; and the target variable.

The fitted scaling objects were saved to ensure that identical preprocessing is applied during model inference, improving reproducibility and consistency.
Feature scaling was applied exclusively to the deep learning model because neural networks are sensitive to differences in variable magnitudes. The ARIMAX model was estimated using the original feature scales.